SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED
CO3: Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.

Assessment Tool Weightage: 10%

QUESTIONS: 50 TOTAL MARKS: 50

Code of Conduct:
I **T Sanhith(192311228)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ______________________________

ANNEXURE A - MCQ

1. A system uses MD5 for integrity checking. Which attack can most likely compromise it?

A. Replay attack

B. Collision attack

C. Phishing attack

D. Brute force on keys

2. Which modification improves security when replacing MD5 in a password system?

A. Use plaintext storage

B. Use SHA-256 with salting

C. Use shorter hash values

D. Disable hashing

3. In a digital signature system, what ensures non-repudiation?

A. Symmetric key

B. Private key of sender

C. Public key of receiver

D. Hash length

4. Which scenario indicates a failure of a hash function?

A. Slow computation

B. Same hash for different inputs

C. Large output size

D. Deterministic output

5. HMAC is preferred over simple hashing because it:

A. Reduces computation time

B. Uses public keys

C. Combines key with hash

D. Eliminates encryption

6. In Kerberos, what prevents replay attacks?

A. Password reuse

B. Timestamp mechanism

C. Hash collisions

D. Key length

7. Which component in Kerberos issues tickets?

A. Client

B. Server

C. KDC

D. Firewall

8. X.509 certificates primarily provide:

A. Data compression

B. Authentication

C. Routing

D. Encryption only

9. What is the main weakness of MD5?

A. Slow hashing

B. Collision vulnerability

C. Large key size

D. Complex implementation

10. SHA-256 is preferred over MD5 because it:

A. Is faster

B. Has shorter output

C. Is collision resistant

D. Uses symmetric keys

11. In digital signatures, hashing is used to:

A. Encrypt message

B. Reduce message size

C. Generate keys

D. Authenticate network

12. Which attack exploits the mathematical principle behind finding hash collisions?

A. Collision attack

B. Dictionary attack

C. Birthday attack

D. Phishing

13. What is required for verifying a digital signature?

A. Sender's private key

B. Receiver's private key

C. Sender's public key

D. Shared secret key

14. In ElGamal signature, security depends on:

A. Hash length

B. Discrete logarithm problem

C. RSA problem

D. AES encryption

15. Schnorr protocol is mainly used for:

A. Encryption

B. Authentication

C. Compression

D. Routing

16. Which property makes it computationally difficult to find two inputs with the same hash?

A. Determinism

B. Collision resistance

C. Speed

D. Length

17. A weak MAC allows:

A. Faster encryption

B. Message tampering

C. Reduced bandwidth

D. Secure authentication

18. Kerberos relies primarily on which cryptographic method?

A. Asymmetric only

B. Symmetric key cryptography

C. Hashing only

D. Quantum cryptography

19. X.509 certificates include:

A. Private key

B. Public key

C. Session key

D. Hash key

20. Which attack exploits the probability of hash collisions?

A. MITM

B. Birthday attack

C. Replay attack

D. DoS

21. Digital Signature Standard (DSS) is based on:

A. AES

B. DSA

C. DES

D. RC4

22. Which ensures integrity in HMAC?

A. Encryption

B. Keyed hashing

C. Compression

D. Encoding

23. In Kerberos, TGT stands for:

A. Ticket Granting Ticket

B. Trusted Gateway Token

C. Temporary Grant Ticket

D. Token Generation Table

24. Which framework uses certificates extensively?

A. Kerberos

B. X.509

C. HMAC

D. MD5

25. Which property is violated if an attacker finds two inputs with the same hash?

A. Pre-image resistance

B. Collision resistance

C. Determinism

D. Efficiency

26. ElGamal signatures are vulnerable if:

A. Key reused

B. Random value reused

C. Hash used

D. Public key shared

27. Which improves MAC security?

A. Removing key

B. Increasing key randomness

C. Using plaintext

D. Reducing hash size

28. Which is NOT a hash property?

A. Deterministic

B. Fixed length

C. Reversible

D. Fast computation

29. Kerberos authentication may fail if:

A. Time not synchronized

B. Hash too long

C. Key too short

D. Network fast

30. X.509 trust depends mainly on:

A. Hash length

B. Certificate authority

C. Encryption speed

D. MAC value

31. Schnorr signatures are considered efficient because of:

A. Smaller signature size

B. Larger keys

C. No hashing

D. No authentication

32. Which is a MAC algorithm?

A. SHA-256

B. HMAC

C. RSA

D. ElGamal

33. Digital signatures provide:

A. Confidentiality only

B. Integrity and authentication

C. Compression

D. Routing

34. Which algorithm is obsolete due to known security flaws?

A. SHA-256

B. MD5

C. HMAC

D. DSS

35. In Kerberos, session keys are used for:

A. Permanent storage

B. Temporary communication

C. Hashing

D. Compression

36. Which ensures sender authenticity in email?

A. Hash only

B. Digital signature

C. Encryption only

D. MAC only

37. Which attack is primarily prevented by HMAC?

A. Replay attack

B. Message tampering

C. DoS

D. Phishing

38. X.509 certificates are validated using:

A. Symmetric keys

B. Public key infrastructure

C. Hash only

D. MAC

39. Which statement is generally true about secure SHA algorithms?

A. Reversible

B. Designed for collision resistance

C. Symmetric

D. Encrypted

40. Which parameter is critical in ElGamal security?

A. Prime number

B. Hash length

C. MAC key

D. Timestamp

41. Which step verifies a digital signature?

A. Encrypt hash

B. Verify signature using sender's public key

C. Generate MAC

D. Compress message

42. Weak hash functions primarily affect:

A. Bandwidth

B. Integrity

C. Routing

D. Compression

43. Kerberos uses which trusted server component?

A. DNS

B. KDC
C. HTTP
D. FTP

44. Which provides integrity and authenticity together?

A. Encryption
B. MAC
C. Routing
D. Compression

45. Schnorr protocol relies on:

A. RSA problem
B. Discrete log problem
C. AES
D. DES

46. Which is vulnerable to practical collision attacks?

A. MD5
B. SHA-512
C. HMAC
D. DSS

47. Which improves digital signature security?

A. Larger key size
B. Smaller hash
C. No hashing
D. Plaintext

48. Which ensures certificate authenticity?

A. MAC
B. CA signature
C. Hash only
D. Encryption

49. What is the main role of a MAC?

A. Confidentiality
B. Authentication and integrity
C. Compression
D. Routing

50. Which combination is most secure for message integrity and authentication?

A. MD5 only
B. SHA + HMAC
C. Plaintext
D. Encryption only